

Chapter 9 — Subject Deep Dives

15 prompts

The rest of the book works across subjects. This chapter does not — each prompt is built around how one subject actually behaves, and what goes wrong when a general tool is pointed at it.

What's in here

Group	Prompts	Use when
Mathematics	3	Word problems, worked examples, error analysis
Science	3	Investigations, controls, lab write-ups
English language arts	3	Close reading, writing tasks, text sets
Social studies	3	Sources, chronology, contested questions
Arts, PE and electives	3	The subjects general tools ignore

Worth knowing before you start

Verify everything factual in this chapter. Historical dates, scientific values, quotations from texts — this is where a confident wrong answer does the most damage, because it looks exactly like a right one.

For science, never accept invented data. Predicted patterns and measured results are different things, and the prompts here keep them separate.

For social studies, ask for the contested question rather than the settled

summary. That is where the thinking is.

For arts and PE, be specific about equipment and space. Generic plans assume a room you do not have.

Every prompt in this chapter

Mathematics

ID	Prompt	Grades
SD-001	Create a concept-first mathematics explanation	3-5, 6-8, 9-12
SD-003	Connect concrete, visual and symbolic mathematics representations	3-5, 6-8, 9-12
SD-004	Build a mathematics error-analysis lesson with verified solutions	3-5, 6-8, 9-12

Science

ID	Prompt	Grades
SD-002	Design an evidence-based science investigation	3-5, 6-8, 9-12
SD-008	Build a claim-evidence-reasoning task from supplied observations	3-5, 6-8, 9-12
SD-009	Audit a science explanation for causation, uncertainty and safety	3-5, 6-8, 9-12

English Language Arts

ID	Prompt	Grades
SD-005	Build a close-reading lesson from an authorized supplied text	3-5, 6-8, 9-12

SD-006	Create an evidence-based writing task with a clear mentor model	3-5, 6-8, 9-12
SD-007	Design a revision lesson that separates ideas, organization and conventions	3-5, 6-8, 9-12

Social Studies

ID	Prompt	Grades
SD-010	Create a source-analysis lesson using origin, purpose and context	3-5, 6-8, 9-12
SD-011	Build a corroboration task with multiple supplied perspectives	3-5, 6-8, 9-12
SD-012	Design a historical claim task that distinguishes evidence from interpretation	3-5, 6-8, 9-12

Arts Physical Education And Electives

ID	Prompt	Grades
SD-013	Create a skill-development lesson with modelling, practice and reflection	3-5, 6-8, 9-12
SD-014	Adapt a performance task for space, equipment and physical access	3-5, 6-8, 9-12
SD-015	Build an observable process rubric for an arts, PE or elective task	3-5, 6-8, 9-12

SD-001 · Create a concept-first mathematics explanation

Grades 3-5, 6-8, 9-12 · Mathematics

WHEN YOU NEED THIS

Students can follow a procedure but need the underlying idea.

COPY FROM HERE

Create a concept-first mathematics explanation.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [FRACTION_EXAMPLES: supply verified information; do not leave blank]
- [REPRESENTATIONS: supply verified information; do not leave blank]
- [PRIOR_KNOWLEDGE: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Connect quantities, diagrams and symbols with equal wholes. Verify comparisons and state the conditions under which the explanation works.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

TO HERE

BEFORE YOU SEND IT

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Example — an editorial illustration, not a logged AI run

Both are one unit fraction short of one. Since $\frac{1}{4}$ is larger than $\frac{1}{6}$, subtracting $\frac{1}{6}$ leaves more: $\frac{5}{6} > \frac{3}{4}$. Confirm with twelfths: $\frac{10}{12} > \frac{9}{12}$. Draw equal 0-1 number lines; different whole sizes would invalidate a direct amount comparison. Transfer: order $\frac{2}{3}$, $\frac{7}{8}$, $\frac{11}{12}$. Answer: $\frac{2}{3} < \frac{7}{8} < \frac{11}{12}$ because their missing unit fractions decrease.

SD-002 · Design an evidence-based science investigation

Grades 3-5, 6-8, 9-12 · Science

WHEN YOU NEED THIS

Students need to investigate a testable question safely.

COPY FROM HERE

Design an evidence-based science investigation.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [APPROVED_INVESTIGATION: supply verified information; do not leave blank]
- [VARIABLES: supply verified information; do not leave blank]
- [SAFETY_LIMITS: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Write controls, repeated observations, units and limitations. Separate predicted patterns from measured results and prohibit fabricated data.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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Example — an editorial illustration, not a logged AI run

Change water temperature and measure time until visible sugar disappears. Keep sugar amount, water volume, cup type and stirring pattern constant. Repeat conditions if time allows. Record actual seconds in a blank table; no data are supplied here. Predicted pattern: warmer water may dissolve sugar faster under these conditions. Limits: judging disappearance and inconsistent stirring. Teacher verifies safe handling and cleanup before use.

SD-003 · Connect concrete, visual and symbolic mathematics representations

Grades 3-5, 6-8, 9-12 · Mathematics

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to connect concrete, visual and symbolic mathematics representations.

COPY FROM HERE

Connect concrete, visual and symbolic mathematics representations.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [CONCEPT: supply verified information; do not leave blank]
- [CONCRETE_MATERIALS: supply verified information; do not leave blank]
- [SYMBOLS: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Explain the mapping between each physical action, diagram element and symbolic operation. State where the analogy breaks down.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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Example — an editorial illustration, not a logged AI run

Concrete model: three groups of two tens and three ones. Visual model: six tens and nine ones. Symbolic model: $3 \times (20 + 3) = 60 + 9 = 69$. Connect each group in the drawing to one addend rather than treating the picture as decoration. Transfer: $14 \times 2 = 28$. The model assumes positive whole-number groups; it does not by itself explain multiplication by a negative number.

SD-004 · Build a mathematics error-analysis lesson with verified solutions

Grades 3-5, 6-8, 9-12 · Mathematics

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to build a mathematics error-analysis lesson with verified solutions.

COPY FROM HERE

Build a mathematics error-analysis lesson with verified solutions.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [VERIFIED_PROBLEMS: supply verified information; do not leave blank]
- [DELIBERATE_ERRORS: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Create clearly fictional wrong solutions with the first invalid step identified. Supply corrected reasoning and a fresh transfer problem.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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Example — an editorial illustration, not a logged AI run

First invalid step: distributing 2 to x but not to 3. Correct: $2x+6=14$; $2x=8$; $x=4$. Check: $2(4+3)=14$. Ask learners why 5.5 fails substitution: $2(5.5+3)=17$. Fresh item: $3(y+2)=18$ gives $y=4$. Label the initial work deliberately incorrect; do not present it as an actual learner record.

SD-005 · Build a close-reading lesson from an authorized supplied text

Grades 3-5, 6-8, 9-12 · English language arts

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to build a close-reading lesson from an authorized supplied text.

COPY FROM HERE

Build a close-reading lesson from an authorized supplied text.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [AUTHORIZED_TEXT: supply verified information; do not leave blank]
- [READING_OBJECTIVE: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Quote only supplied passages and anchor questions to them. Distinguish text evidence from reasonable interpretation.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-006 · Create an evidence-based writing task with a clear mentor model

Grades 3-5, 6-8, 9-12 · English language arts

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to create an evidence-based writing task with a clear mentor model.

COPY FROM HERE

Create an evidence-based writing task with a clear mentor model.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [WRITING_PURPOSE: supply verified information; do not leave blank]
- [SOURCE_EVIDENCE: supply verified information; do not leave blank]
- [MENTOR_TEXT: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Build a writing task and annotated mentor model using authorized or original text. Do not invent quotations, authors or research findings.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-007 · Design a revision lesson that separates ideas, organization and conventions

Grades 3-5, 6-8, 9-12 · English language arts

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to design a revision lesson that separates ideas, organization and conventions.

COPY FROM HERE

Design a revision lesson that separates ideas, organization and conventions.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [WRITING_SAMPLE: supply verified information; do not leave blank]
- [REVISION_PRIORITY: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Separate idea development, organization and conventions. Focus revision on one priority before surface editing.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-008 · Build a claim-evidence-reasoning task from supplied observations

Grades 3-5, 6-8, 9-12 · Science

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to build a claim-evidence-reasoning task from supplied observations.

COPY FROM HERE

Build a claim-evidence-reasoning task from supplied observations.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [ACTUAL_OBSERVATIONS: supply verified information; do not leave blank]
- [QUESTION: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Build a claim-evidence-reasoning task with uncertainty and alternative explanations. Never strengthen a claim beyond the supplied data.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-009 · Audit a science explanation for causation, uncertainty and safety

Grades 3-5, 6-8, 9-12 · Science

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to audit a science explanation for causation, uncertainty and safety.

COPY FROM HERE

Audit a science explanation for causation, uncertainty and safety.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [SCIENCE_EXPLANATION: supply verified information; do not leave blank]
- [VERIFIED_SOURCE: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Check mechanism, correlation versus causation, uncertainty and procedure safety. Cite the supplied source location for corrections where possible.

Output:

Return an issue table with location, evidence, severity and smallest correction; then the corrected artifact and unresolved decisions.

TO HERE

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SD-010 · Create a source-analysis lesson using origin, purpose and context

Grades 3-5, 6-8, 9-12 · Social studies

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to create a source-analysis lesson using origin, purpose and context.

COPY FROM HERE

Create a source-analysis lesson using origin, purpose and context.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [SOURCE_TEXT: supply verified information; do not leave blank]
- [KNOWN_ORIGIN: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Analyze origin, purpose, context and limits. Mark unknown provenance explicitly and avoid filling gaps from stereotypes.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

TO HERE

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SD-011 · Build a corroboration task with multiple supplied perspectives

Grades 3-5, 6-8, 9-12 · Social studies

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to build a corroboration task with multiple supplied perspectives.

COPY FROM HERE

Build a corroboration task with multiple supplied perspectives.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [TWO_OR_MORE_SOURCES: supply verified information; do not leave blank]
- [INQUIRY_QUESTION: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Compare agreement, contradiction and independence of sources. Do not treat multiple copies of one account as independent corroboration.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-012 · Design a historical claim task that distinguishes evidence from interpretation

Grades 3-5, 6-8, 9-12 · Social studies

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to design a historical claim task that distinguishes evidence from interpretation.

COPY FROM HERE

Design a historical claim task that distinguishes evidence from interpretation.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [HISTORICAL_QUESTION: supply verified information; do not leave blank]
- [AUTHORIZED_SOURCES: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Separate supported claims, interpretations and unresolved questions. Avoid invented quotations, dates or false equivalence among evidence quality.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-013 · Create a skill-development lesson with modelling, practice and reflection

Grades 3-5, 6-8, 9-12 · Arts / PE / electives

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to create a skill-development lesson with modelling, practice and reflection.

COPY FROM HERE

Create a skill-development lesson with modelling, practice and reflection.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [PERFORMANCE_SKILL: supply verified information; do not leave blank]
- [APPROVED_EQUIPMENT: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Sequence model, focused practice and reflection with observable criteria. Avoid medical or physical-training advice beyond teacher-approved activities.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-014 · Adapt a performance task for space, equipment and physical access

Grades 3-5, 6-8, 9-12 · Arts / PE / electives

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to adapt a performance task for space, equipment and physical access.

COPY FROM HERE

Adapt a performance task for space, equipment and physical access.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [TASK: supply verified information; do not leave blank]
- [SPACE: supply verified information; do not leave blank]
- [EQUIPMENT: supply verified information; do not leave blank]
- [ACCESS_CONSTRAINTS: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Adapt performance access while preserving the learning goal. Offer equivalent evidence modes without inferring diagnoses.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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SD-015 · Build an observable process rubric for an arts, PE or elective task

Grades 3-5, 6-8, 9-12 · Arts / PE / electives

WHEN YOU NEED THIS

Choose this focused tool when your immediate task is to build an observable process rubric for an arts, PE or elective task.

COPY FROM HERE

Build an observable process rubric for an arts, PE or elective task.

Required inputs:

- [GRADE_SUBJECT: exact age/grade and subject]
- [GOAL: the learning or communication purpose]
- [PROCESS_OBJECTIVE: supply verified information; do not leave blank]
- [OBSERVABLE_EVIDENCE: supply verified information; do not leave blank]
- [CONSTRAINTS: duration, format, resources and approved access requirements; write none only if confirmed]

Specific requirements:

Create a rubric for process, technique and reflection. Do not score body type, innate talent or expensive equipment.

Output:

Return the named artifact with all questions, directions, examples or text needed for the task; put teacher keys and notes separately from student/family-facing text.

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